

Performance Measurement using Fuzzy System Perspective and Soft System Methodology

New Idea of Business Process Management

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Abstract—Soft System Methodology (SSM) is a systematic approach for tackling real-world of Business Process Management (BPM). There are some disadvantages of using Soft System Methodology (SSM): reliance on an expert, the result become fuzzy and has separated indicator with hard system. Now, we are discussing new idea that standardizes a Key Performance Indicator for decision making on the Soft System Methodology (SSM) assisted by Fuzzy Question in System Perspective.

Keywords—component; soft system methodology, key performance indicator, fuzzy logic

I. INTRODUCTION

Business Process Management has been a knowledge area of information system and technology. There are always a set of process running in a system. ABPMP [1] concludes the maturity of an information system depends on business process stability. Soft system approach is one of major problem solving strategy to solve the ill-defined problem on business process modeling. The ill-defined problem [2] is identified by several different perceptions towards the problem conditions. Soft system approach is used to simplify a gap between current and desirable conditions of business process. However, Soft System Methodology (SSM) can be a comprehensive solution that has two points of view: real situation and system approach.

Process transformation is a final step of SSM analysis. But SSM is too “soft”. In a process of building the system, handling structured problems is also important [4]. The hard systems view is a vision that the world is composed of many systems and the performance of these systems is optimized by a semantic procedure [4]. For example, a company A has production process P and being analyzed by a Soft System Methodology to decide a root definition and propose process transformation for P to become X according to Y measurement to fulfill Z purposes. There is some P structured component in the whole system which not included when deciding X, Y and Z. In this case the “hard” system of P is technology.

Above describe that soft system perspective is one of the effective methodology for process analysis. Next section, this will discuss about background and motivation of this research.

II. BACKGROUND AND MOTIVATION

A. Soft System Methodology Characteristic

Soft System Methodology has seven stages of problem solving strategy. The stages are grouped into real world strand and system thinking strand as illustrated in Fig 1.

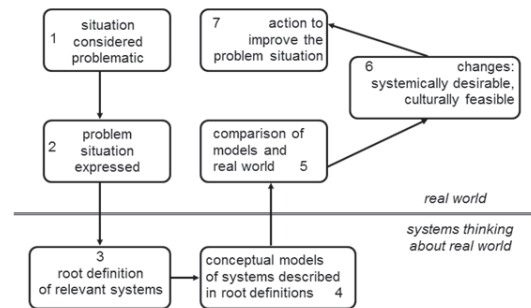


Fig.1 Seven Stages SSM

Those seven stages have to be done by an expert out of the system [2]. Step 1 is to identify a gap happened in business process. Collect data with interview and document analysis. Step 2 creates a rich picture of whole activity model for business process. Step 3 identifies the root definition of relevant system with CATWOE analysis. Step 4 describes conceptual model of system that define input/output of transformation to become X according to Y measurement to fulfill Z purposes. Step 5 compares of model (system thinking) and real world with three criteria: efficacy, efficiency and effectiveness. Step 6 concludes changes, systematically desirable and culturally feasibility. At last step 7 defines set of actions to improve the problem situation.

It has its own characteristic. SSM sees a whole system as a soft, not exactly hard. It analyzed like common people sense. It has holistic view for all business process, easy to practice and can be the best solution for the most common business process situation. It's also focused to how the problem defined well to propose a process transformation.

Although soft system solves so much common problem situation, soft system is unstructured and cannot be quantified. The methodology didn't mention how system builds. Expert factor and soft characteristic makes SSM become fuzzy so that solution. It also depends on the expert who involved in that case for that time. To accompany this, SSM need hard system factors that quantified conceptual model.

B. Proses Performance Measurement

At the last of SSM stages, performance measurement is needed to decide transformation process according to models. This performance measurement prevents overstated and overestimated organization to the business process. Process performance contains of time: cycle time, cost: resource cost and opportunity cost, capacity: transaction and throughput, quality: satisfaction, variation and error/ defect rate [1].

Performance measurement must have list of point that indicates each smallest part of business process. It's called Key Performance Indicator (KPI). KPI must contain these characteristic: alignment, accountability, predictive, actionable, few in number, easy to understand, balanced/linked, transformative, standardized, context-driven, reinforced and relevant [1].

There are some related works using performance measurement in clinical process modeling [3] and business process analysis using performance measurement [5]. Both of those relate works are using KPI.

C. Fuzzy Logic Theory

During the performance measurement we used to make evaluation with complex problem and relatively fuzzy. Such fuzzy words in SSM as "very satisfied", "satisfied", "quite satisfied" and "not satisfied" can be adopted as mathematical expressions. Based on fuzzy mathematics theory, fuzzy comprehensive measurement model is an evaluation method to take fuzzy process to the measuring objects, through making fuzzy transformation and weighting distribution to the quantitative data and fuzzy standard [6].

Theoretically, fuzzy logic deals with reasoning that is approximate rather than fixed logic if compared with traditional binary logical set. Let see binary logic P with expression

$$P = [\text{true} \mid \text{false}] \quad (1)$$

For fuzzy logic:

$$P_{\text{true}} = (0 \dots 1) \text{ and } P_{\text{false}} = (0 \dots 1) \quad (2)$$

Fuzzy logic uses two continuous value for true value and false value. See the example in Fig. 2.

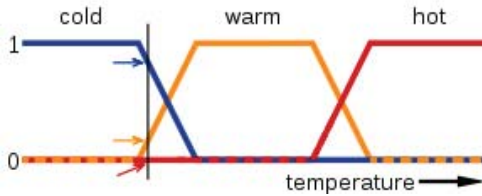


Fig.2 Temperature in Fuzzy Logic

Fuzzy logic has a flow to measure and predicting values. It includes fuzzification, qualitative modeling, qualitative simulation and defuzzification.

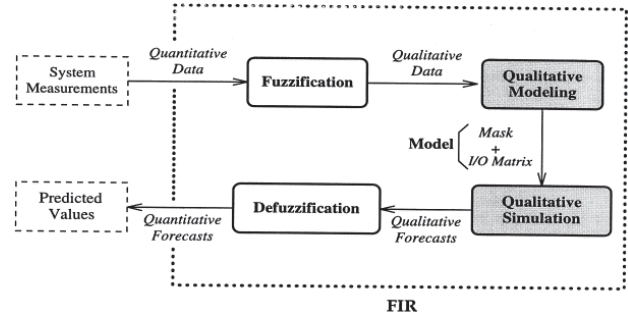


Fig.3 Fuzzy Logic Flow

What is the correlation of fuzzy logic and SSM? Fuzzy logic is used to measure the Key Performance Indicator in System Perspective. Unfortunately, system perspective and structured hard system in SSM is being handled by an expert. An expert judgment can be subjective and fuzzy logic is one of proposed solution for this.

1) Fuzzification

The fuzzification comprises the process of transforming crisp values into grades of membership of fuzzy sets. For example, there is a KPI "fast response helpdesk". KPI value is fuzzy and the value is one of fuzzy set A.

$$A \in \{\text{quick, normal, don't care}\} \quad (3)$$

To measure that indicator we can use quantitative statistic. Let see 75 people state the helpdesk is quick, 20 people state normal response and 5 people don't care about helpdesk. We have the result helpdesk response is 85% quick. How to measure that?

For each KPI, if it's quick the value is 1, if it's normal the value is 0.5 and if it's don't care the value is 0. The result is $75 \times 1 + 20 \times 0.5 + 5 \times 0 = 85\%$ quick. This step called fuzzification.

2) Qualitative Inference

Next step weighting and leveling step is used to make a conclusion and categorize the fuzzy result. There are other linguistic variable of fuzzy set B. Fuzzy set B is the result of qualitative inference process based on defined rules.

$$B \in \{\text{fast, normal, slow}\} \quad (4)$$

The rule in this SSM case is used to categorize data. For example if it has categories: slow [0, 30), normal [30, 80) and fast [80, 100] so the value 85% is categorized to fast. This step called qualitative inference.

3) Defuzzification

Defuzzification is performed according to the membership function of the output variable. The purpose is to obtain a crisp result value.

III. THE SOLUTION AND RESULT

Generally, Soft System Methodology consists of 7 stages. Former considering problem stage and take solution action is the later. Based on research problem, in order to avoid dependencies on an expert analysis an accurate measurement needs for determine process performance measurement in Soft System Methodology. There are some of suggestions in order to accomplish Soft System Methodology: Key Performance Indicator measurement could be involved when the real world and system thinking compared. This KPI has two points of views both soft and hard system perspective; Fuzzy Question is used to determine indicator reasoning. It's not just efficacy, efficiency and effectiveness but also the fuzzy predicted value results.

The new extended SSM could be created based on Soft System Methodology characteristics, the possibility to add "hard" system factors in SSM, process performance measurement using KPI and fuzzy logic theory. So the seven stages become extended illustrated in Fig. 4.

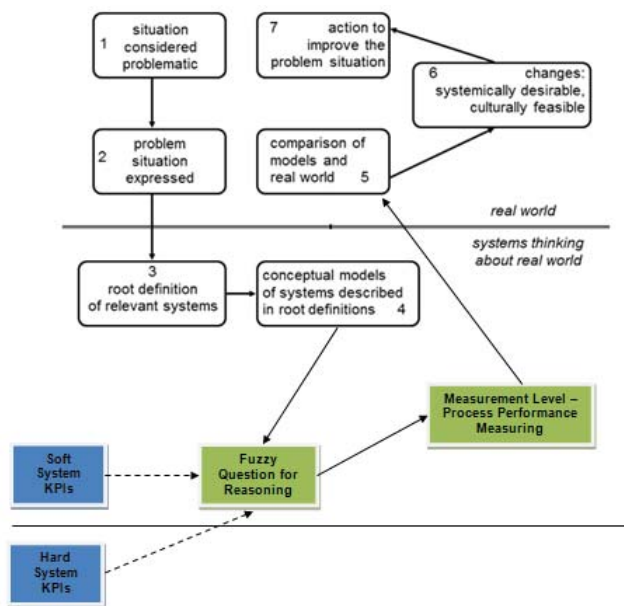


Fig.4 Extended SSM

Fig. 4 shows comprehensive measurement of business process using Key Performance Indicator both from soft system and hard system. Measurement method is using fuzzy logic and gets the qualitative level result from quantitative data. This measurement method is needed when the model and real problem compared.

IV. CASE STUDY

A new theory must be validated with the test. To test the extended SSM above, this is a business process example. Business process of prescription queue contains of:

- 1) *Patient comes to the clinics.*
- 2) *Patient get queue for doctor diagnosis or for asking prescription.*

- 3) *Doctor writing the prescription.*
- 4) *Prescription is given to patient.*
- 5) *Patient goes to suggested pharmacy.*
- 6) *Patient gives the prescription to pharmacy front office.*
- 7) *Patient waits while medicine prepared.*
- 8) *Patient called.*
- 9) *Patient gets medicine and gives payment.*

As an expert we create an extended SSM based on cases above.

A. Situation Considered Problematic

Problem:

- Dissatisfaction indicated from patients who complain about ineffective queue process.
- Prescription registration in pharmacy's front office and doctor is written manually and take too much time.

B. Problem Situation Expressed

Clinics always use manually and written prescription. After doctors diagnose patient they give a written prescription in related paper template. This prescription paper is taken manually to suggested pharmacy. So we can calculate that "waiting time" from whole process is T.

T = waiting time for doctor prescription creation + travel time of prescription from doctor to pharmacist + waiting time for pharmacy's front office jobs + waiting time for pharmacist preparing medicine and transaction. The more waiting time could slow a whole business process.

C. Root Definition of Relevant Systems

Root definition: transformation from manual prescription to digital and consider efficiency factor of waiting time to accelerate business process.

X: prescription transformation manual to digital

Y: waiting time efficiency

Z: accelerate the process

CATWOE analysis:

C: patient

A: doctor, front officer, pharmacist

T: prescription transformation manual to digital

W: prescription writing efficiency

O: clinic owner

E: medical policies

D. Conceptual Models of Systems Described



Fig.5 Activity Diagram

E. Identifying Key Performance Indicator

Although there are so many indicators in this business process, the KPI is confined to one only KPI. It's writing prescription indicator. This KPI is observed from both soft and hard system.

TABLE I. KEY PERFORMANCE INDICATOR

KPIs: fast of prescription process			
Performance Assessment	Input	Output	Outcome
Variable: Time T Factors (t ₁ -t ₅): Doctor thinking time about prescription, writing prescription time by doctor, travel time clinic to pharmacy, front-office queuing time and pharmacist preparing time	Time	Time	Fast response of the system

F. Fuzzy Question for Reasoning

In this step an expert have to collect real data from hard system. The weight of term quick is 1, weight 0.5 for normal and weight 0 for don't care/ no answer. Statistically the collected fuzzy data from patient tells:

- t₁: 2 quick, 7 normal, 10 don't care (28.95%)
- t₂: 3 quick, 1 normal, 15 don't care (18.42%)
- t₃: 5 quick, 2 normal, 12 don't care (31.58%)
- t₄: 1 quick, 2 normal, 16 don't care (10.53%)
- t₅: 0 quick, 2 normal, 17 don't care (5.26%)

TABLE II. FUZZY MEASUREMENT

Indicator	Average Value	Factor (t ₁ -t ₅)	Value
Prescription process	18.94%	Doctor thinking time about prescription	28.95% quick
		Writing prescription time by doctor	18.42% quick
		Travel time clinic to pharmacy	31.58% quick
		Front-office queuing time	10.53% quick
		Pharmacist preparing time	5.26% quick

The fuzzification process makes the result of time (T) 18.94% quick for KPI: fast of prescription process from doctor to pharmacist.

G. Process Performance Measuring

Next step the expert decide the rule. This rule is used to inference the result of linguistic term. In this case the simple rule inference transform {quick, normal, don't care} term to {fast, average, slow} term.

TABLE III. FUZZY MEASUREMENT LEVEL

Measurement Level	Slow	Average	Fast
Prescription Process	T > 3 hours [0, 20] %	T 45'' – 3 hours (20, 80] %	T < 45'' (80, 100] %

Just because the result of fuzzification is 18.94% quick, based on the rule the waiting time is still **slow**.

H. Comparison of Models and Real World

This is regular SSM step, compares of model/ system thinking with real world with three criteria. The result of fuzzy logic process improves the level of confidence for an expert to state the effectiveness statement.

TABLE IV. MODEL COMPARISON

Activity	Efficacy	Efficiency	Effectiveness
Prescription Process	Prescription obtained	Prescription input twice, by doctor and pharmacist	Waiting time is still slow
	efficacy reached	efficiency reached	effectiveness isn't reached

I. Changes, Systematically Desirable, Culturally Feasible

During the performance measurement we used to make evaluation with complex problem and relatively fuzzy. Such few in number, easy to understand, balanced/linked.

J. Action to Improve the Problem Situation

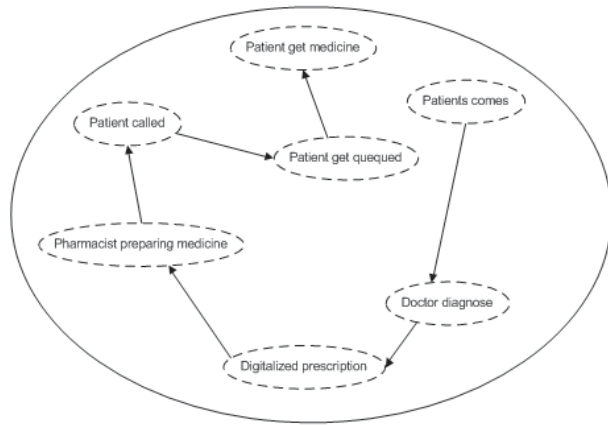


Fig.6 Activity Model

This extended SSM exactly measure each indicator so that an expert can decide the best action solution of the problem and also decide to execute that action or not.

V. CONCLUSION AND FUTURE DEVELOPMENT

The result of discussion in this paper we can conclude that First, the idea of extended SSM become a standard for decision making in SSM using KPI. Key Performance Indicator could be used comprehensively to create performance measurement in SSM. Second, the dependencies to an expert in SSM could be reduced by using KPI that involve hard system from related business process. Last, fuzzy logic helped this extended SSM to measure the unstructured problem in comparing models with real problem and increase level of confidence for an expert in that comparison stage.

This extended SSM using KPI and fuzzy question could reduce disadvantages of general SSM. The methodology shows how the system builds. The transformation proposal is come from statistical quantitative data in the very first the quantitative. The KPI and fuzzy question is also reducing the subjectivity factor of an expert because it is blended with hard system.

The fuzzy system perspective is like a bridge-hybrid between hard system and soft system. Note that the fuzzification, inference and defuzzification process is not as same as how fuzzy machine works. The main idea of this extended SSM is the fuzzification process, the KPI is helped system to create rule of inference process and defuzzification process is optional.

Further, this extended SSM could be a new area for future research about relation of Key Performance Indicator with efficacy, effectiveness and efficiency.

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